

Arrays and Linked Lists are two fundamental data structures used to store collections of elements. An array stores elements in contiguous memory locations, enabling constant-time access using an index. This makes arrays highly efficient for reading and updating elements when the index is known. However, arrays have a fixed size in many languages, meaning resizing requires allocating new memory and copying elements, which is costly.

Linked Lists store elements as nodes, where each node contains data and a reference to the next node. Unlike arrays, linked lists do not require contiguous memory and can grow or shrink dynamically. Insertion and deletion operations are efficient because they only require pointer updates. However, accessing an element requires traversing the list from the beginning, resulting in linear-time access.

Arrays are best suited for scenarios requiring fast access and minimal insertions or deletions, such as lookup tables. Linked lists are preferred when frequent insertions and deletions are required, such as implementing stacks and queues. Both structures have trade-offs, and choosing between them depends on the specific application requirements.

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